Ship Auto Way Maps Points Tracking

Dashboard PySide6 untuk monitoring telemetry kapal model secara real-time, perencanaan waypoint, dan pengiriman konfigurasi ke Remote via User-Side ESP32.

File utama

File	Deskripsi
`Local Monitor Dashboard-beta1.1.1.py`	Versi aktif (beta 1.1.1)
`Local Monitor Dashboard-beta1.1.py`	Versi sebelumnya

Arsitektur

```
Dashboard --USB serial--> User-Side ESP32 --ESP-NOW--> Remote-Side (kapal)
```

Tab aplikasi

Live Data

- Peta tracking posisi kapal
- Indikator live (yaw, heading, rudder, RPM, baterai, speed, WP index)
- Time-series plots
- Logging CSV (nilai tampilan, bukan raw)
- Setup koreksi rudder sensor
- Tampilan route Home → WP (setelah Send to Remote)

Map Points

- Klik peta untuk menambah waypoint
- **Set Home Point** — Home dari koordinat serial terbaru
- Panel **Send to Remote** — kirim WP + tuning + verifikasi read-back
- Panel **Setup** (di bawah Send to Remote) — dialog algoritma auto + tuning Alg 1
- Tabel waypoint + garis route biru di peta
- Snapshot waypoint ke folder `WayPoints/` (CSV)

Analyze

- Load CSV log (format display / raw / legacy)

- Peta + 3 plot + panel indikator
- Slider timeline untuk replay data

Protokol serial (ke User-Side)

Kirim waypoint

```
$WPSET,<home_lat>,<home_lon>,<wp_count>,<lat1>,<lon1>,...
```

Kirim tuning

```
$TUNSET,<alg>[,kp,kd,arrive_m,rudder_max]
```

Default alg **2** (stub). Alg 1 membutuhkan 4 parameter.

Read-back tuning

```
$TUNGET
```

Respons yang di-handle dashboard

Pola	Makna
`$WACK,OK,WP`	Waypoint diterima Remote
`$WACK,OK,TUN`	Tuning disimpan NVS
`$WACK,ERR,...`	Error WP/TUN
`$TACK,<alg>,...`	Read-back sukses
`$TACK,ERR,...`	Read-back gagal

Alur Send to Remote

1. Validasi: connected, Home set, minimal Home + 2 WP
2. `$WPSET` → `$WACK,OK,WP`
3. `$TUNSET` → `$WACK,OK,TUN`
4. `$TUNGET` → bandingkan `$TACK` dengan Setup
5. Status: **Verified OK** / **MISMATCH** / **TIMEOUT**

Timeout per langkah: **3 detik**.

Telemetry masuk (23 kolom)

Raw fixed-point dari User-Side; dashboard menampilkan nilai fisik:

Kolom	Skala tampilan
timestamp, lat, lon	asli
speedMps, yaw, heading, rudder, servo, accel, gyro, rpm, battery	÷ 100
distance_to_wp	÷ 10
track_wp_index, mode_auto	integer

Setup dialog (Map Points)

- Radio **Alg 1** (waypoint + PD) / **Alg 2** (stub, default)
- Spinbox Alg 1: Kp, Kd, WP arrive (m), Rudder max (°)
- **Read from Remote** — `$TUNGET`, update UI dari `$TACK`

Menjalankan

```
cd Pythonfile/Way_Points_Tracking python "Local Monitor Dashboard-beta1.1.1.py"
```

Dependencies

- Python 3.10+
- PySide6
- pyserial
- PyQtWebEngine (peta)

Folder output

- `WayPoints/` — snapshot CSV waypoint saat Send to Remote
- Log CSV — lokasi sesuai pengaturan tab Live

Catatan

- Pastikan MAC ESP-NOW Remote ↔ User-Side cocok sebelum uji lapangan
- Connect serial ke port User-Side (115200 baud)

- CH6 RC ≥ 1750 untuk mode Auto di kapal

Author

Chandra P — Ship Model Control System

Versi

- **beta 1.1.1** — waypoint send, tuning NVS, verify TACK, Analyze tab